

Factory assembled magnet system					
Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

Section 2a

Mechanical installation (Part 1)

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1. CHECK OF THE MAGNET SHOCK-INDICATORS

Shock indicators must be checked before unloading the magnet from the truck in order to define responsibilities in case of damage to the magnet

If one or more shock indicators are tripped, the magnet did withstand higher mechanical forces than it was designed for. This means there is a risk of internal (invisible) magnet damage.

NOTE

- *When you accept a magnet, which has a shock-indicator, tripped: make a note on the waybill to be countersigned by the transporter.*
- *Only shock indicators mounted on the magnet, not on the magnet crate, are valid for this procedure*

Remove all shock indicators from the magnet.

If cold spots (condensation) are visible on the surface (not on the turret) call the Helpdesk MR before moving the magnet into the hospital.

It is important to check a.s.a.p. whether the magnet is damaged or not. Access to the room should be maintained in the event the magnet has to be replaced. Following information has to be collected and sent to the Helpdesk MR in Best.

- Which shock-indicator(s) (position / direction / 10 and/or 15 G) are tripped?
Remark: If 5G indicators are tripped, no actions are required. Only make a note on the waybill.
- Does the cryostat have any visible damage? If so, make pictures including pictures of the packing.
- Copy of the waybill, including the notes you have made on the waybill.
- 12 planes bare plot, including a description of the iron near the magnet.
- Helium boil-off figures.

These measurements will be compared with the data before shipment in order to advise you about the actions to be taken.

Possible actions are:

- The magnet performance is still within specification. Continue with the installation
- The magnet can be locally repaired.
- The magnet has to be replaced.

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Ad a:

In the factory twelve shock indicators are mounted on the magnet.

There are two kind of shock-indicators:

1. Type 1 indicator is built up of four metal balls and two springs. When not tripped, each spring presses two balls (one at each side) against the housing. One spring is mounted in horizontal direction and one in vertical direction.

As soon as the shock indicator (and the magnet) is accelerated faster than specified in a certain direction, the balls fall out of their position. In this situation the shock indicator has been tripped.

2. Type 2 indicator has an arrow that turns blue in one of the two directions when tripped.

No actions are required if only type 1 shock-indicators are tripped

Ad e:

"Boil-off" is the evaporation of liquid helium inside the He-vessel.

Two methods can be used to measure the boil off:

1. Make two readings with an interval of several days. Divide the difference of these two readings by the time (in hours) between these two readings.

$$\text{Boil off} = [\text{l/h}] \quad \frac{(\text{Reading at } t1 [\text{l}] - t2 [\text{l}])}{t2 - t1 [\text{hours}]}$$

2. Connect a helium calibrated flowmeter to the gas exhaust unit, using the necessary reducing pieces and a hose (inner diameter > 5 mm, length > 5 m). Do a leak test with snoop (supplied with the system) to make sure all the gas is coming trough the flowmeter. Let the pressure stabilize for 12 hours, do not change the magnetic field and avoid cryogenic actions. After 12 hours or longer measure the gas-flow. Multiply the gas-flow (litre/minute) by 0.090 to get the liquid helium consumption in litre/hour. The most accurate way is to take the mean value over a period of one week to take into account variation of boil off due to changes in atmospheric pressure. (2 readouts every day). The boil-off figures (< 1.0 l/h with refrigerator system switched off). are specified as an average over 5 days with gradients NOT pulsing.

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2. TRANSPORTING THE MAGNET

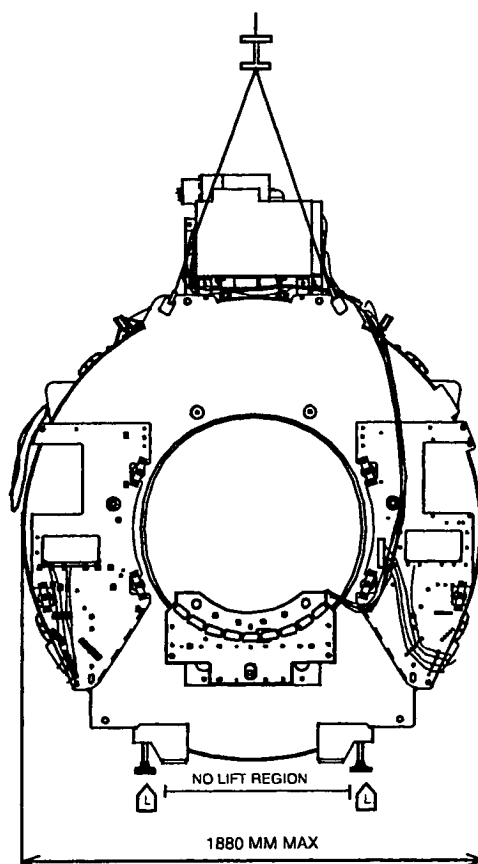
Magnet rotation tool:

If the transport route height is limited, it might be needed to rotate the magnet using a "Magnet Rotation Tool".

The "Magnet rotation tool" is available via the commercial logistic department. For information regarding the transport dimensions when using the "Magnet rotation tool" refer to the Planning Reference Book. See also Service Newsletter MR111. A user manual is supplied with the "Magnet Rotation Tool".

The magnet may only be lifted using the lifting eyes located on top of the magnet or by lifting it at the designated areas on the bottom of the magnet. Lifting cable angle and points must be in accordance with the markings. For further information refer to the "Shipping, handling & storage" document supplied with the magnet.

Figure 1- Magnet transport

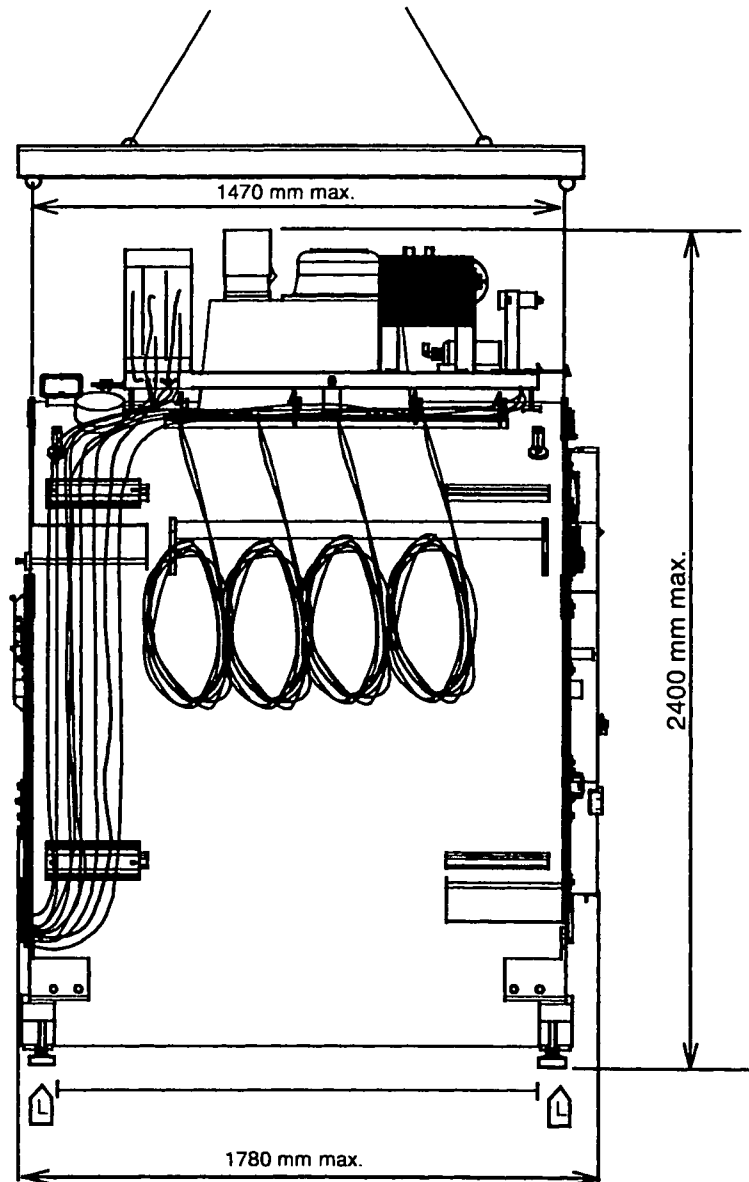


Weight and dimensions (Factory assembled magnet system)

	Standard and Omni gradients	Power and Master gradients
Height	2400 mm	2400 mm
Length	1780 mm	1780 mm
Width	1880 mm	1880 mm
Weight 1.5T	4200 Kg	4460 Kg
Weight 1.0 T	4000 Kg	4260 Kg
Weight 0.5 T	3800 Kg	N/A.

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Figure 2 - Magnet transport



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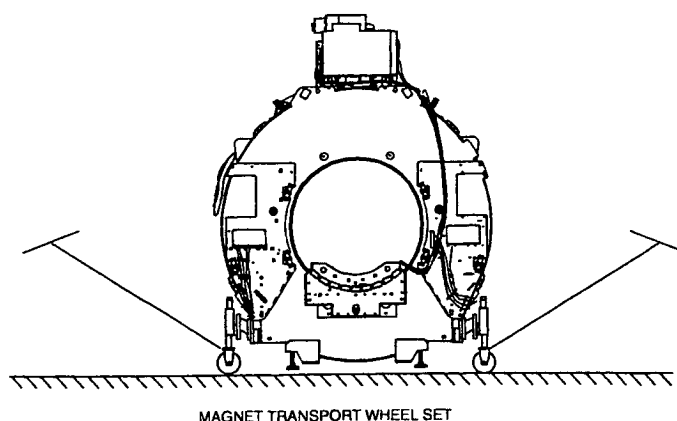
CAUTION

The "magnet transport set T5" may not be used for Intera magnets. The extensions of this set are too weak. Instead use "magnet transport set T5/NT/Intera" (see Section 1 Service Tools).

Cover the floor with metal plates to prevent the floor from being damaged.

A set of 4 caster wheels can be mounted to the magnet for transporting it into the RF room.

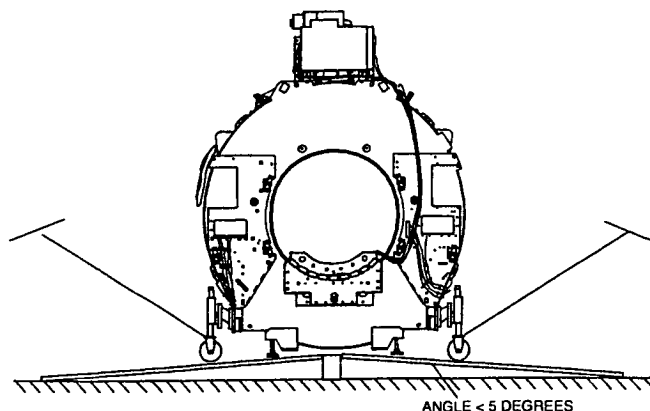
Figure 3 - Wheel set for magnet transport



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If a ramp is used to transport the magnet over a threshold, the angle of the ramp may not be more than 5 degrees. Otherwise the magnet will touch the ramp. The magnet can be pushed over the ramp with four or five people. Make sure that the turret of the magnet does not touch the RF cage.

Figure 4 - Magnet transport over a ramp



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3. TRANSPORT OTHER SYSTEM PARTS

For transporting the gradient coil combination, the forks of the pallet truck must have a minimal length of 150 cm.

When cabinets are damaged, make a note on the waybill, to be countersigned by the transporter. Contact your local Customer Support manager to discuss further steps. If possible make photos.

3.1. CHECK OF THE COPLEY GRADIENT AMPLIFIER SHOCK-INDICATORS

Shock indicators must be checked after unpacking in order to define responsibilities in case of damage to the Copley gradient amplifier.

If one or more shock indicators are tripped, the amplifier did withstand higher mechanical forces than it was designed for. This means there is a risk of internal (invisible) damage inside the cabinet.

There are two kind of shock indicators used:

- One type of indicator has an arrow that turns blue in one or two directions when tripped. These Shock indicators can be seen from the rear side of the cabinet.
- The other shock indicators have a small horizontal tube, which turns red when tripped. They are on the top cover inside the Copley cabinet.

NOTE

When you accept a Copley gradient amplifier, which has a shock-indicator, tripped: make a note on the waybill, to be countersigned by the transporter. If possible make photos.

In case of tripped shock indicators the following information has to be collected and sent to the Helpdesk MR in Best:

- The system reference number
- Copley serial number (At the front upper right corner of the Copley).
- Which shock-indicator(s) (position / direction / 15 G and/or 25 G and/or 50 G) are tripped?
- Does the gradient amplifier cabinet has any visible damage? If so, make photos including photos of the packing.
- Copy of the waybill, including the notes you have made on the waybill.

If there is no visible damage, you can continue the installation.

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4. POSITIONING THE MAGNET

Due to the fact that threaded holes for the patient support are already present in the floor pads, it is needed to position the magnet with respect to these holes. For accurate positioning it is required to position the magnet using the supplied magnet-positioning jig.

The magnet must be placed on the four pads or on a floor which has sufficient surface strength. No fixation to the floor is needed, unless required by local regulations.

In earthquake areas special precautions must be taken for mounting the magnet to the floor (for details refer to the Planning Reference Book). The mounting must be done in such a way, that the magnet remains insulated from the RF cage

The distance between the rear magnet interface plate to the finished wall or, in case something is placed on that wall, to the protruding part must be at least 760 mm.

4.1. TOOLS

- Crowbar (length ≥ 1.5 metre)
- Magnet transport set
- Spirit level, length 500 mm, accuracy 0.1 mm/m
- Spirit level (3 m transparent water hose)
- Ruler
- Two open-ended spanners, size 30
- Digital Volt meter
- Magnet positioning Jig (Supplied with system)

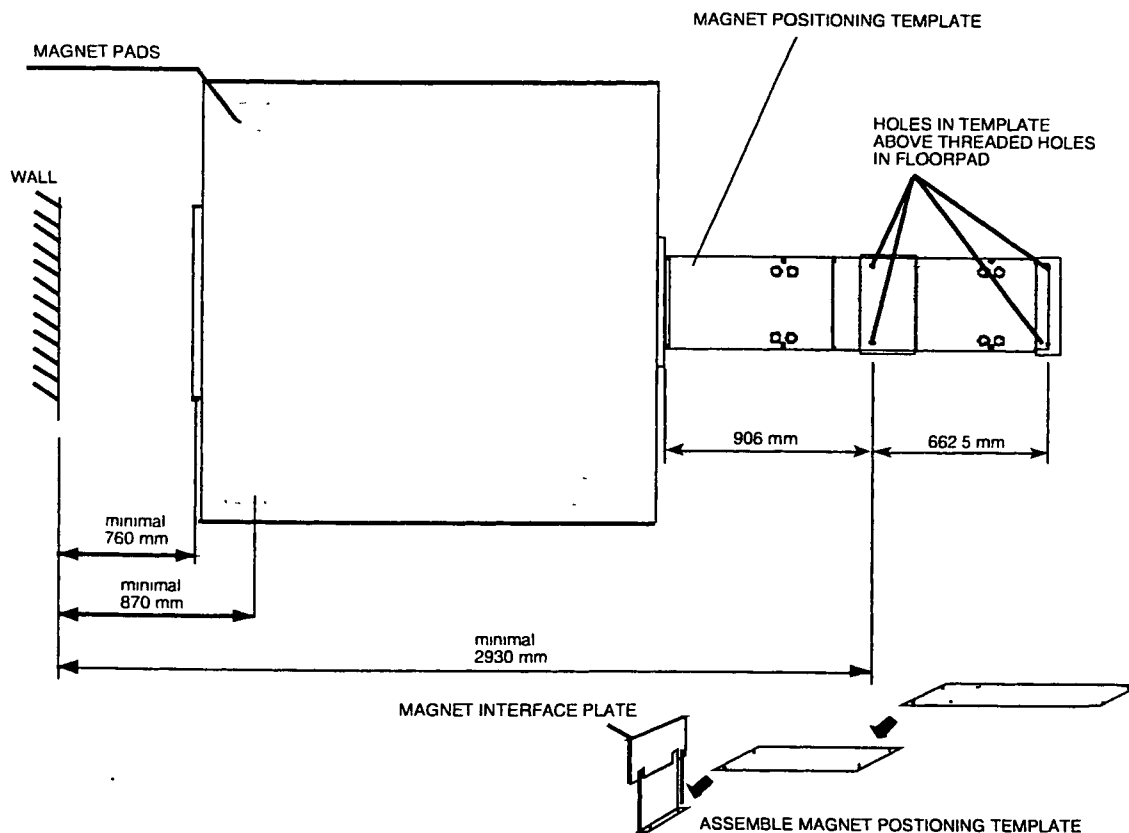
4.2. PROCEDURE

Check that the distance between the centre-point of the two rear magnet pads to the rear wall, or in case something is placed on the wall, to the protruding part is ≥ 870 mm. If no pads are present, measure the distance between the patient support fixation points (closest to the rear RF wall) to the rear wall, or in case something is placed on the wall, to the protruding part is ≥ 2930 mm.

1. Mount the four magnet feet underneath the magnet. The magnet feet are supplied with the magnet in a separate box.
2. Place the magnet in position using the magnet transport wheels.
3. Mount the magnet-positioning jig to the magnet.
4. Position the magnet (one-way valve towards the rear) so that the four holes in the magnet-positioning jig are positioned above the threaded holes in the floor pads. Position the magnet as good as possible using the transport wheels. For exact positioning loosen the wheels and position the magnet using the crowbar.

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Figure 5 - Magnet position

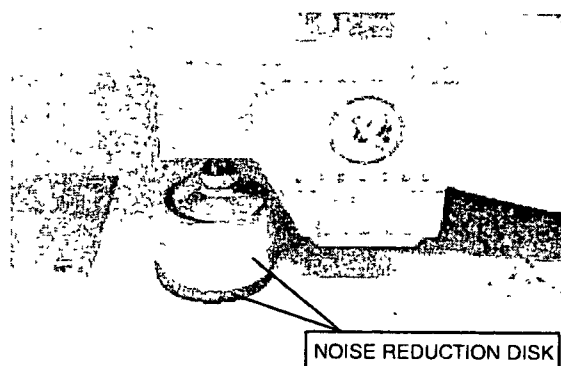


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After placing the noise reduction disks underneath the magnet, the weight of the magnet must be spread equally between the four disks. If the magnet weight is spread between two disks too long (more than 1 hour), the disks might distort.

- If the magnet is in the correct position, place one noise reduction disk underneath each foot.

Figure 6- Noise reduction disk



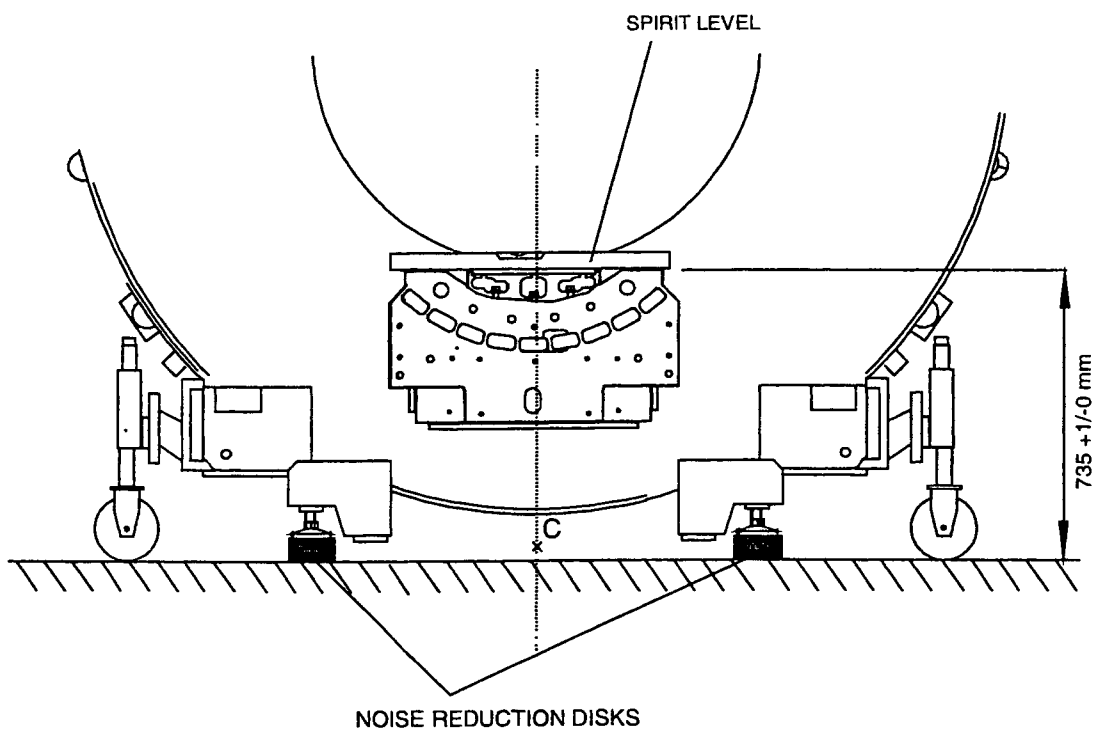
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5. ADJUST THE HEIGHT OF THE MAGNET

Height adjustment and leveling can be done by using the four adjustable feet of the magnet. Use the magnet transport wheels to lift the magnet.

Adjust the height to: distance between the top of the gradient carrier and the finished floor: 735 ± 1.0 mm (only at the front side).

Figure 7 - Height adjustment of the magnet

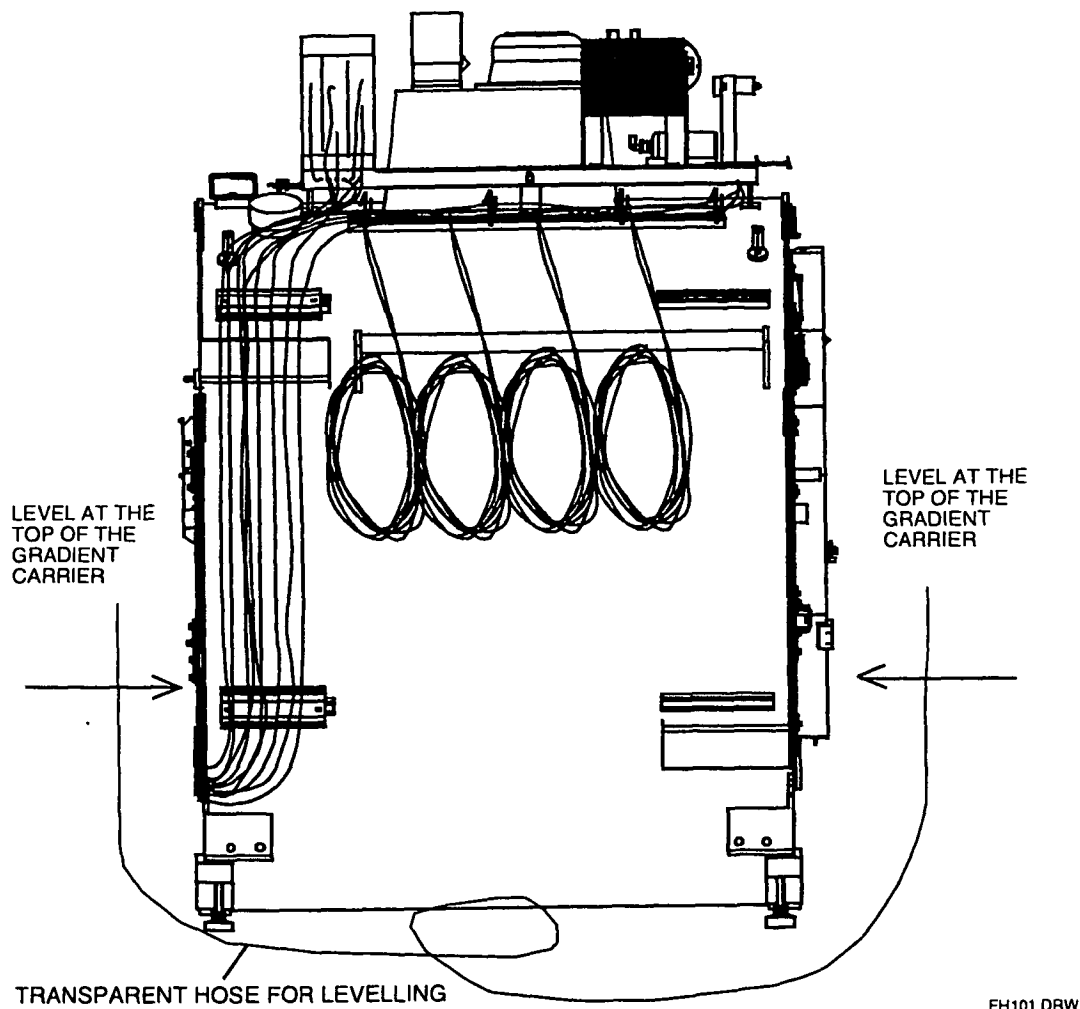


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6. LEVEL THE MAGNET

Figure 8 - Longitudinal leveling of the magnet



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1. Only at the front of the magnet, level the magnet perpendicularly (side ways) using a spirit level. Apply equal force to the nuts of all feet.
2. Level the magnet in the longitudinal direction, use a transparent hose filled with water. Measure at the top of the gradient carrier front and rear. Apply equal force to the nuts of the two rear feet.
3. Lift the wheels, until the wheels are free from the floor. Check with the digital voltmeter the resistance between the magnet and the RF cage. $R \geq 200 \text{ Kohl}$. Correct insulation, if not within limits. Write the measured value in the MRL.
4. Lock the adjustments made by fixing the nuts on top and bottom of the threaded rods of the magnet feet. Use two spanners of 30 mm.
5. Remove wheels.

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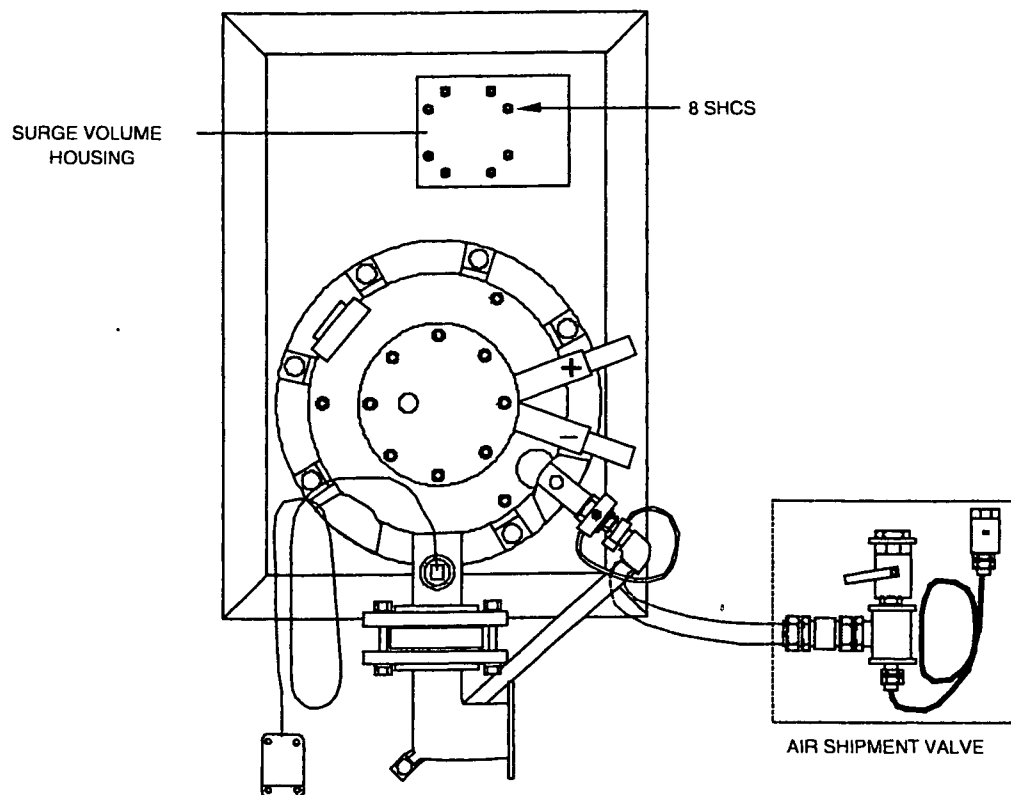
7. REPLACE THE AIR SHIPMENT VALVE

To prevent air from entering the magnet during air transport, an air shipment valve has been mounted to the turret.

Procedure for the helium exhaust (see Figure 9):

1. Release the magnet pressure by opening the valve on the air shipment valve.
2. Disconnect the air pressure release valve.
3. Mount the flexible helium exhaust hose to the fixed helium exhaust valve assembly
4. Close the ball valve (yellow valve).
5. Return the air shipment valve to Best.

Figure 9 - Helium exhaust type 4 with air shipment valve



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8. MOUNTING THE GAS EXHAUST LINES

A gas exhaust line for inside the RF enclosure is supplied with the system. The RF feedtrough is supplied together with the pre-delivery kit to be mounted by the RF cage supplier. The external exhaust pipes should also be present.

Mount the remaining parts of the gas exhaust line according to the following procedure.
Refer to drawing Z3-4.5 of the drawings manual

CAUTION

Take care that the gas exhaust line does not touch other conductors in its neighborhood (e.g. false ceiling). This may cause the so-called 'spike' artifact.

WARNING

Never cut the flexible exhaust pipe in order to adapt its length. In that case the connection between the flexible pipe and its flange is too weak so that you remain with an unsafe situation. The glue that is present between pipe and flange is essential for its strength.

1. Mount the flexible hose (500-mm) to the RF feedtrough interface (detail B of drawing Z3-4.5).
2. Mount the other flexible hose (1500-mm) to the exhaust of the magnet (detail D of drawing Z3-4.5).
3. Mount the hooks to the ceiling by screwing them onto the M5 bolts prepared by the RF supplier (detail F). The distance between two hooks should be between 500 and 800 mm).
4. Mount the pipes between the two flexible hoses. Depending of the distance use the 1000-mm pipes and/or the 500-mm pipe. To mount the flexible hose to a pipe see detail E. To mount two pipes together see detail C. Mount the pipes to the ceiling (detail A-A and detail F of drawing Z3-4.5).
5. Mount the gutter to the gas exhaust line with the strings (detail A-A of drawing Z3-4.5). The drip-tray may not touch the RF feedtrough and must be mounted so that water in this drip-tray will run to the magnet. (In case of a quench, the liquid air is caught in the drip-tray and will drip on the magnet outer housing, where it evaporates).
6. Fasten all bolts.

9. CLOSING THE MAGNET TRANSPORT PASSAGE

The transport opening may be closed when all large and heavy items have been brought into the examination room. As the passage forms part of the shielded RF room (examination room), proceed with the utmost care. The cage builder's instructions have to be followed to ensure a properly shielded room.